QXU6033 Advanced Polymer Chemistry

2025B | Prof. Yamin Abdouni and Prof. Sulafudin Vukusic

■ Contents

1. Introduction to Module

■ Assessment

Activity	Score
Professional Practice Evaluation(supervision)	20%
Coursework	30%
Final Exam	50%

1.1. Study Goal

■ Controlled Radical Polymerisation

Mechanistic principles of controlled / living radical polymerisation (CRP)
Kinetic control, equilibrium processes, and radical deactivation strategies
Molecular weight control, dispersity (Đ), and end-group fidelity

■ Core CRP methodologies

Nitroxide-Mediated Polymerisation (NMP)
Reversible Addition–Fragmentation Chain Transfer (RAFT)
Atom Transfer Radical Polymerisation (ATRP)

■ Advanced macromolecular architectures

Self-Condensing Vinyl Polymerisation (SCVP)
Strathclyde synthesis and hyperbranched structures
Star polymers and multi-arm architectures

■ From synthesis to structure

Block copolymer design and synthesis
Microphase separation and self-assembly

Introductory Discussions I

Q: A has a structure that enables it to polymerise through chain-growth polymerisation, while B is step-growth.
Scenario 1: Imagine you start polymerising A, and half-way through the polymerisation, you add C, also another monomer, which can polymerise through chain-growth polymerisation, and, that can polymerise together well with A (this is called "copolymerise"). Separately, imagine Scenario 2: Use B instead A, then add D, a monomer, which can polymerise through step-growth polymerisation, and, that can copolymerise well with B. At the end of both polymerisation processes, what type of polymer samples will you have in each Scenario?
A: S1: AAAAA-(random A and C)-(if one is finish, pure another one).
S2:(block B)-some D-(block B) and cycle B.

2. Controlled Radical Polymerisation

Note
CRP, controlled / living radical polymerisation.
Also Reversible Deactivation Radical Polymerisation, RDRP.

2.1. Towards Living Polymerisation

As long as "living" polymer is kept tightly closed (no H_2O, ROH, CO_2, O_2) more monomer can be added and the molecular weight will increase.

1. MW increases steadily with time/conversion
2. Narrow PDI(< 1.2 or 1.3)
3. Can add more monomer or another monomer/blocks.

Kinetic Issue

To make FRP "living", need to inhibit termination and chain-transfer without affecting initiation/propagation too much.

Strategy 1: Reversible Termination

Also, Reversible Deactivation.

$$\sim P_n \cdot + X \cdot \underset{\mathrm{k}_a}{\overset{\mathrm{k}_d}{\rightleftharpoons}} \sim P_n - X \qquad \text{("Dormant(sleep) FR")}$$

Note: Do not use Strategy 1, either refer to specific chemistry (NMP, or TEMPO) or "reversible deactivation"

Strategy 2: Catalytic/Assisted Reversible Termination

$$\sim P_n \cdot + X - Y \cdot \underset{\mathrm{k}_a}{\overset{\mathrm{k}_d}{\rightleftharpoons}} \sim P_n - X + Y \qquad \text{("Dormant(sleep) FR")}$$

X can be single atom transferred to $P\cdot$, Y normally a transition metal.
Also, **Catalytic Reversible Deactivation.**

Reversible Chain-Transfer(Degenerative Transfer)

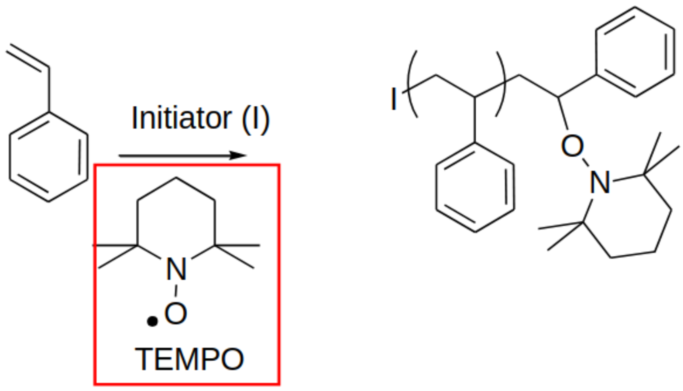
$$\sim P_n \cdot + \sim P_m - X \underset{\mathrm{k}_a}{\overset{\mathrm{k}_d}{\rightleftharpoons}} \sim P_n - X + \sim P_m \cdot$$

In processes $\sim P_n \cdot$ or $\sim P_m \cdot$ themselves, **Termination still occurs**.

To be effective
Reversible Termination should be $k_d > k_a$
Reversible Chain-Transfer is $k_d \sim k_a$

2.2. Controlled Radical Polymerisation by Reversible Deactivation

Specific chemistries:
Nitroxyl radicals (e.g. TEMPO) for Nitroxyl Mediated Polymerisation (NMP)



TEMPO: (2,2,6,6-Tetramethylpiperidin-1-yl)oxyl radical

Note
The equilibrium with TEMPO doesn't happen necessarily at every single monomer addition.

2.3. Controlled Radical Polymerisation by Catalytic Reversible Deactivation

Specific chemistries:

■ ATRP (Atom-transfer free radical polymerisation)

A bromo (or chloro) compound + Cu^+ complex act as the initiator: does not require an external free-radical initiation system.